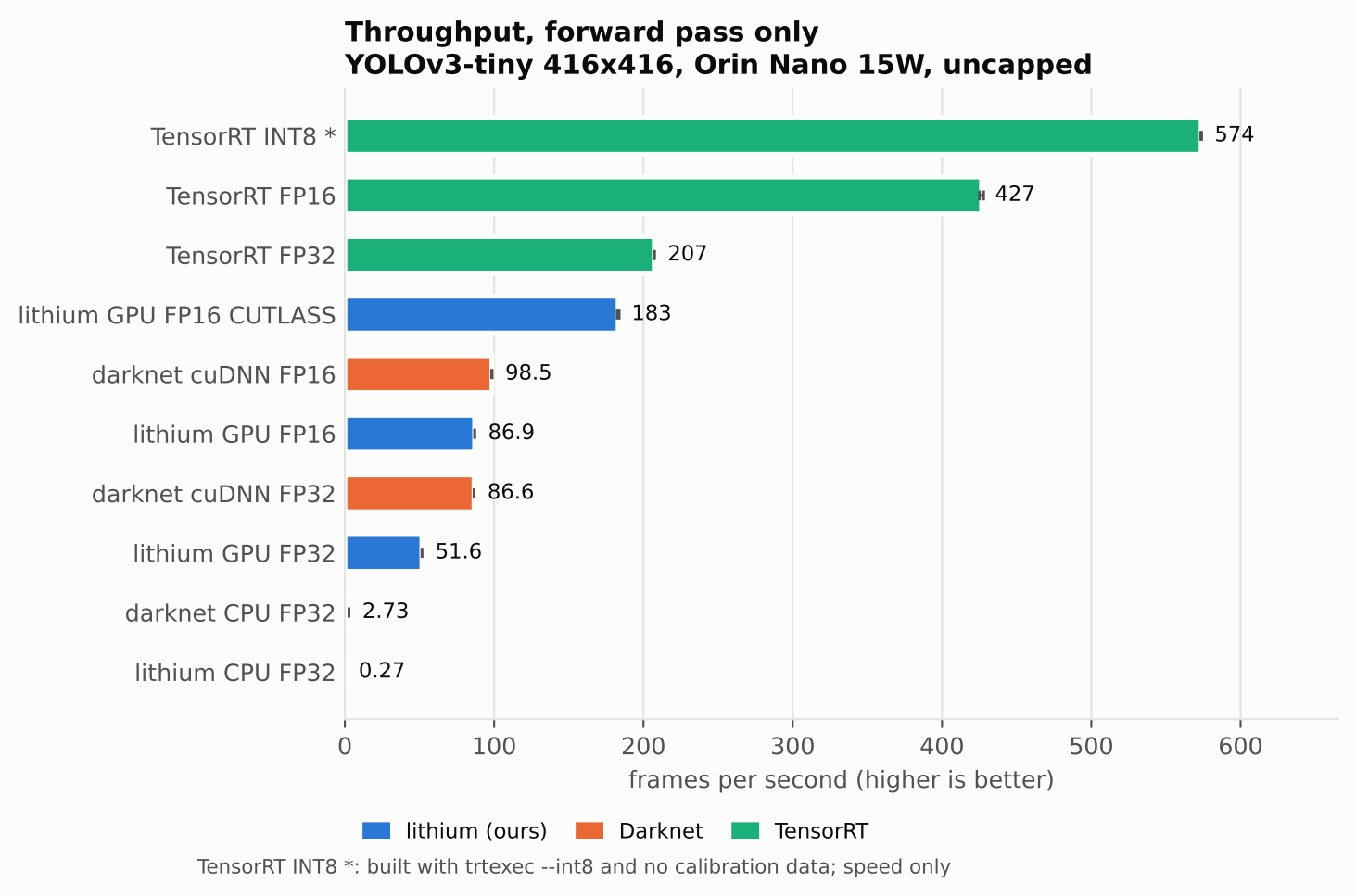
Throughput, forward pass only
YOLOv3-tiny 416x416, Orin Nano 15W, uncapped
TensorRT INT8 *
TensorRT FP16
TensorRT FP32
lithium GPU FP16 CUTLASS
darknet cuDNN FP16
lithium GPU FP16
darknet cuDNN FP32
lithium GPU FP32
darknet CPU FP32
lithium CPU FP32
574
427
207
183
98.5
86.9
86.6
51.6
2.73
0.27
0
100
200
300
400
500
600
frames per second (higher is better)
lithium (ours)
Darknet
TensorRT
TensorRT INT8 *: built with trtexec --int8 and no calibration data; speed only